

Trait Anxiety and Impaired Response Inhibition: Go/NoGo ERP study

Response Inhibition Mediated by Trait Anxiety in a Go/NoGo with ERP Neural Correlates
for Performance
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King's Collage London

Author Note

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Response Inhibition Mediated by Trait Anxiety in a Go/NoGo with ERP Neural Correlates for Performance

Eysenck's attentional control theory (Eysenck et al., 2007) suggests that at least part of the dysfunctional neural circuitry in highly anxious people overlaps with inhibition control. The medial prefrontal cortex (mPFC), including the anterior cingulate cortex (ACC), seems to function as a hub for complex self-regulating cognitions including memory, impulse control and decision making (Holroyd & Yeung, 2012).

The mPFC as a higher order region requires dense connections to various other brain regions, in order to facilitate executive functions such as directing goal-oriented behaviors, monitoring performance and processing errors in an output independent way (Holroyd et al., 1998). This connection is highly conserved across multiple species indicating its pivotal role in sustaining life by allowing potentially life-threatening stimuli to automatically be brought into conscious awareness where it can be evaluated and responded to promptly (Yao et al., 2018).

Irregular mPFC activity correlates with amygdala hyperactivity in people with anxiety disorders (Monk et al., 2008), which in turn correlates to the severity of symptoms and the degree cognitive impairments as found in recent research (Sehlmeier et al., 2010).

Anxiety disorders are also linked to distinct cognitive traits such as negativity bias where the adverse aspect of a stimuli is consistently exaggerated and impairments such as increased reaction time which has been demonstrated in a Go/No-Go paradigm. Furthermore, research indicates an interacting model where mPFC function is affected by anxiety and vice versa (van Noordt & Segalowitz, 2012),

Behavioral performance in the Go/No-Go task seems to exclusively impair the inhibition component of cognition since research shows a lower response accuracy and an increased response latency in people with high trait anxiety (HTA) compared to low trait anxiety (LTA) in NoGo trials that is not seen in the Go trials (Xia et al., 2020), these behavioral measures of impaired inhibitory control has been shown to correlate with neural measures.

The frontal-midline N2 which peaks 200-400ms after the stimuli is thought to represent motor planning and inhibition; the P3 peaks 450-600ms post-stimuli and is thought to represent the monitoring and evaluation of an inhibition (Raud et al., 2020). The NoGo-N2 in HTA shows a reduced peak amplitude and an increased latency compared to LTA, similarly the NoGo-P3 in HTA has an increased peak amplitude and latency compared to LTA. The differences between HTA and LTA in NoGo trials was not conserved in Go trials which reflects the behavioral results. This suggests that the neural basis for perceptual distortions seen in anxiety specifically effect the way a negative stimulus is perceived, in a way that weakens the potency of the inhibitory signal leading to errors, and in response to these errors the P3 shows a stronger response. This provides evidence to the theory of a global negativity bias in anxiety and links the perceptual phenomenon directly to functional deficits which can explain daily difficulties the condition brings like impaired concentration, negative anticipation, and irrational fears.

In this paper we seek to provide evidence for this theory using EEG recordings and behavioral measures using a Go/No-Go paradigm from students scored on anxiety using the

STAI-T. The high temporal resolution that comes along with electrophysiological recordings would allow for us to analyze the various sub-processes involved in the task.

Methods

Design and Participants

A total of 64 second-year undergraduate Neuroscience and Psychology students from King's Collage London participated in this study. The study consisted of three experiments, a Multisensory Emotion Task (MET, $n = 34$), Resting state ($n = 64$) and a Go/No-Go task ($n = 30$). Each of these experiments utilized an EEG design to collect data on the underlying neural activity during the tasks or at rest. No demographic data was collected however age can be assumed to range from 19-25, class gender ratio as 64% female from a diverse set of ethnic and cultural backgrounds (Guerra, 2021).

All participants were required to complete 4 questionnaires: Barret's impulsivity Scale, Cognitive Failures Questionnaire (CFQ), Interpersonal reactivity Index (IRI) and State Trait Anxiety Index (STAI). Each designed to test personality and behavioral traits associated with cognitive functions relating to impulse control, lapses in memory and perceptual functions, dispositional empathy and situational/trait anxiety score respectively.

This paper focuses exclusively on the Go/No-Go experiment as a way of measuring the underlying neural processes involved in inhibition and cognitive control, and to what degree individual variations in trait anxiety can impairs this process. The use of EEG allows for us to analyze the sub-processes (N2 and P3) associated with inhibitory cognitive control under a high degree of temporal resolution. ERP analysis was preferred over Oscillation analysis since it offers the ability to investigate the task related events.

Procedure

The data collection phase lasted 5 weeks, each week had 2 testing days, and each testing day had 3 separate testing sessions where 1 student was measured each session for a total of 30 participants over 5 weeks. 30 out of the total 64 students were assigned to the Go/No-Go study.

Students were read their instructions, asked to confirm understanding, and give oral consent. A 64 electrode EEG cap was placed on the students at the start of the experiment, and they sat Infront of a 120hz monitor. The task had 5 blocks of 100 trials, in 85% of the trial's participants were instructed to respond upon the presentation of the Go stimuli, which we selected as the number '2', and abstain from responding when presented with the No-Go stimuli 'Z'. Stimuli was presented in a pseudorandom order with 425 total Go trials and 75 total No-Go trials. Each stimulus was presented for 100ms and followed by a 750ms response window. A fixation cross was presented on the screen for the duration of the response window and the inter-trial period. At the end of the 500 trials, 10-minute resting state EEG recordings were collected for the 3rd experiment if this study, where participants were instructed to looks the fixation cross, relax their mind and try not to do any heavy cognitive processing. At the end of the 5-week data collection period all participants, from both MET and Go/No-Go experiments, were followed up and asked to complete 4 questionnaires: The Barret's impulsivity, Cognitive failure, IRI (reactivity index) and STAI (state anxiety).

EEG recording

The 64 channel EEG recordings were collected through a BioSemi ActiveTwo AD box set up with a battery. Reference electrodes were placed above, below, left, and right of the eyes with the addition of the 2 mastoids either side of the ears. Neural activity was recorded at a sample

rate of 1000Hz referentially to the average of the CPz and Pz then re-referenced off-line to the average of the left and right mastoid. Signal was digitized and input into the recording computer using a BioSemi USB convertor box. Recordings were collected through the ActiView software running on an iMac (27-inch, 2017), after data consistency was clarified on the same software and corrected if needed. EEG recordings were synced up to the triggers ran on the MATLAB experiment file and saved at the end of the experiment.

Processing

Data was processed on MATLAB (v9.11, 2021) using the EEGLAB plugin. EEG data was down sampled to 256Hz. An online low-pass filter of 80 Hz was applied by the software. Data was then put through a high-pass filter 0.1Hz, and a low-pass filter of 30Hz was applied as supported by literature (Xia et al., 2020; Funderud et al., 2012).

EEG data was then epoched to 200ms prior and 800ms after the onset of the stimulus with the whole epoch mean being used for the baseline correction, epochs were binned under correct Go and NoGo trials.

Gross artefacts over $\pm 40 \mu V$ were flagged for removal then data was visually inspected, and excessively noisy epochs were excluded. On average each participant had 16 ± 8 epochs removed, with an average of 484 epochs being retained for analysis per participant.

For each participant pre-processed data was put through an independent component analysis (ICA) in order to separate the components and localise their origins. Individual components were visually analysed. Any components classified as artifacts with an ICA confidence score above 90% were automatically removed, any components above a 60% threshold were flagged and left for visual evolution. This process allowed for the isolation and removal of eye blink components and other anomalous artifacts.

ICA weights were conserved for the purpose of creating averages relative to each datapoints size. N2 stimuli-locked ERPs were generated by averaging the frontocentral electrodes F1, Fz, F2, FC1, FCz and FC2 as supported by previous literature (Xia et al., 2020; (Xia et al., 2020; Sehlmeier et al., 2010). Grand averaged ERPs were generated for HTA and LTA conditions in both Go and NoGo trials. Statistics describing the averaged ERP's peak amplitude and peak latency were extracted for each condition and trial for later analysis.

Analysis

Statistical analysis was performed in R-Studio(v4.1) with descriptive statistics being presented as mean and standard deviation (SD). An alpha below 0.05 was considered significant. Multiple two-way ANOVAs were computed using measures of behavioural (Accuracy and RT) and ERP characteristics (N2/P3 peak amplitude/latency) as the dependent variables, with trial (Go/NoGo) as the within-subject variable and trait anxiety (HTA or LTA) as the between-subject variable. P values were presented as adjusted for multiple comparisons using Bonferroni method.

Results

Behavior

A total of 30 students had their EEG data collected, from which 9 had to be excluded due to incomplete STAI-T questionnaires, 20 participants data was retained for analysis. Low/high trait anxiety (LTA/HTA) was classified as the lower and upper 25% of the STAI-T. HTA participants had reduced response accuracy in NoGo trials but not Go trials whilst maintaining a significantly longer response latency across all trials compared to LTA, suggesting that the effect anxiety has on cognition is specific to inhibition in this task, but that there is also another overarching effect causing global reduction in cognitive flexibility and therefore overall latency. Table 1 shows summary of group statistics and behavioural measures.

Accuracy. A significant interaction between TA and trial on response accuracy ($F(1,8) = 6.16$; $p < .05$; $\eta^2p = 0.345$). A simple main effect analysis found that TA had a significant effect on ACC in NoGo trials ($F(1,8) = 12.3$; $p < .01$; $\eta^2p = 0.016$) but not Go trials ($F(1,8) = 0.537$; $p = .56$; $\eta^2p = 0.043$). A Bonferroni corrected Pairwise comparison further investigating this main effect found response accuracy in the HTA group ($97 \pm 6\%$) to be significantly lower compared to LTA ($98 \pm 1\%$) in NoGo trials ($p < .01$), but not Go trials ($p = .57$). A summary of behavioural data is shown in Figure 1.

Table 1

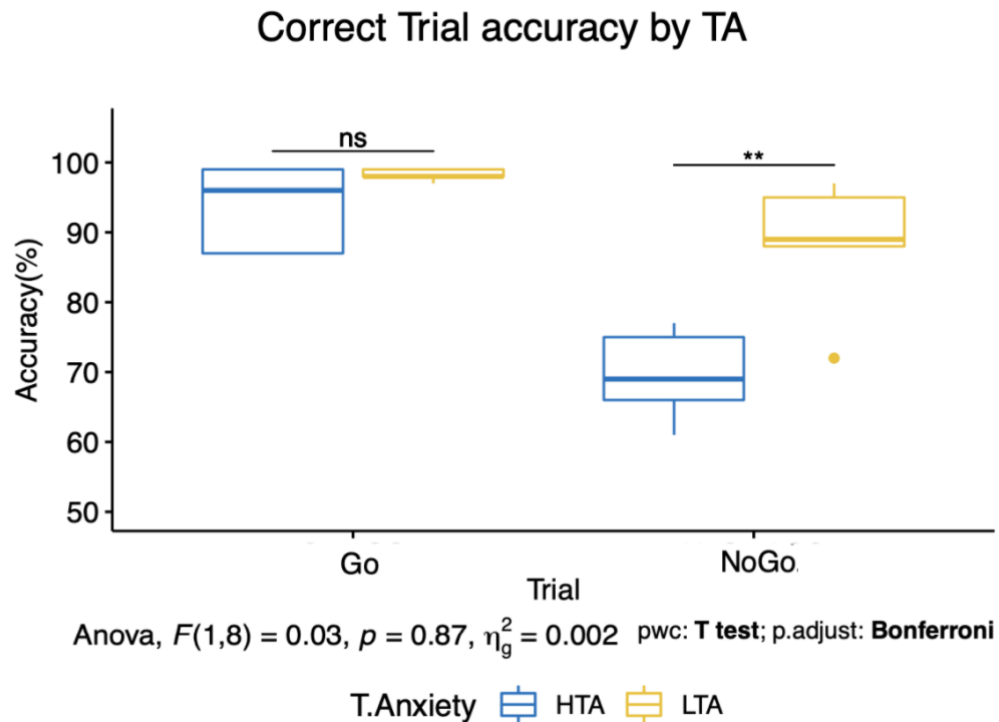
Means and standard deviations for ACC and RT

TA	Go		NoGo	
	ACC (%) <i>M(SD)</i>	RT (ms) <i>M(SD)</i>	ACC (%) <i>M(SD)</i>	RT (ms) <i>M(SD)</i>
HTA	97(6)	371.5(41.4)	70(6)	312.2(30)
LTA	98(1)	379.6(37)	88(10)	242(68)

Note. M and SD represent mean and standard deviation, respectively. Accuracy as ACC given in percentage correct responses and reaction time RT given in ms.

Figure 1

Reaction time and Accuracy by in Go/NoGo trials by TA group



note. significance given as the results of a pairwise Bonferroni corrected t-test.

ERP analysis

Data describing the peak amplitude latency of the front medial N2 and P3 was extracted from grand-averaged ERPs from the two TA groups. The N2 is defined as the second negative going peak which was found to be stimulus locked to 250-350ms post presentation whereas the P3 is found to peak between 350-450ms post stimuli. Figure 2 illustrates the ERP and ANOVA results.

N2 Peak Amplitude

The N2 peak amplitude interaction of trial by TA was not significant ($F(1,8) = 0.009$; $p = .925$; $\eta^2_p = 0.001$). There was also no effect of either TA ($F(1,8) = 1.8$; $p = .216$; $\eta^2_p = .09$) or trial ($F(1,8) = 1.3$; $p = .28$; $\eta^2_p = 0.084$) on the averaged N2 peak amplitude.

N2 Peak Latency

The N2 peak latency interaction of trial by TA was not significant ($F(1,8) = 0.792$; $p = .39$; $\eta^2_p = 0.044$). There was also no effect of either TA ($F(1,8) = 0.435$; $p = .528$; $\eta^2_p = .029$) or trial ($F(1,8) = 0.97$; $p = .358$; $\eta^2_p = 0.054$) on the averaged N2 peak amplitude.

P3 Peak Amplitude

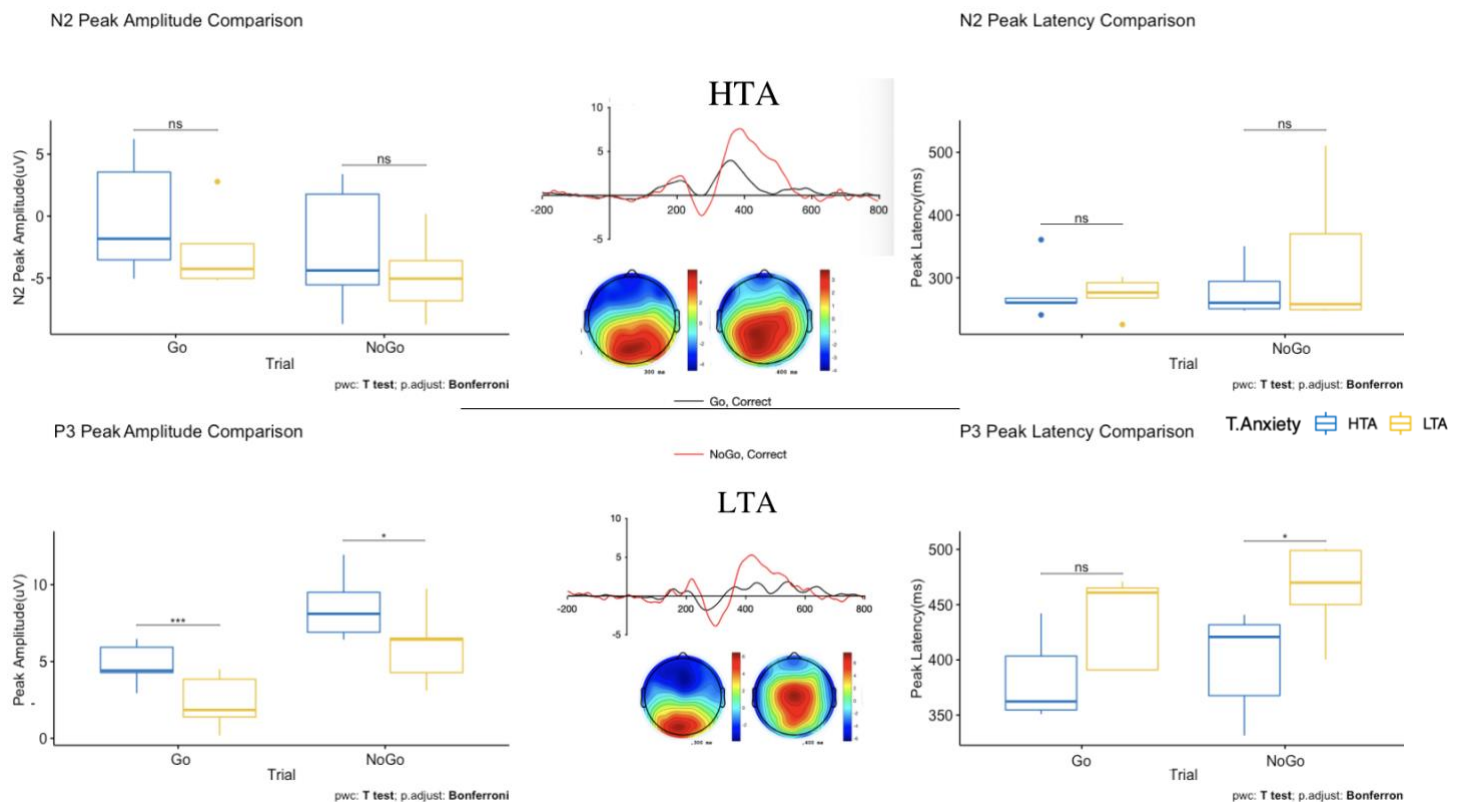
The P3 peak amplitude interaction of trial by TA was not significant ($F(1,8) = 0.004$; $p = .95$; $\eta^2_p = 0.00021$). However upon inspecting the main effects, TA ($F(1,8) = 7.29$; $p < .05$; $\eta^2_p = 0.32$) and Trial ($F(1,8) = 17.26$; $p < .01$; $\eta^2_p = 0.51$) had a significant effect on the P3 peak amplitude. A pairwise comparison reveals a significant difference in P3 amplitudes between HTA and LTA in Go trials ($t(4) = 13.4$, $p < .001$) and NoGo trials ($t(4) = 11.6$, $p < .05$).

P3 Peak Latency

The P3 peak latency interaction of trials by TF was not significant ($F(1,8) = 0.13$; $p = .72$; $\eta^2p = 0.007$), however a main effect of TA on P3 latency was found ($F(1,8) = 8.27$; $p < .05$; $\eta^2p = 0.37$). A pairwise comparison revealed a significant difference in the P3 latency between HTA and LTA groups in NoGo trials ($t(4) = -3.32$, $p < .05$) that was not present in Go trials ($t(4) = -1.8$, $p = .147$).

Figure 2

Visualisation of ERP waves, Topographic Scalp Map and ANOVA results



note. box plots represent data from ANOVA models investigating the impact trait anxiety has on the amplitude and latency of N2 and P3 ERP components. Results show Bonferroni corrected p values.

Discussion

The effect trait anxiety has on cognitive inhibitory control is not well established. Here we implemented a Go/NoGo task design to evaluate the differences between HTA and LTA students in behavioral measures of cognitive control in relation to neural components relevant stimuli appraisal and response inhibition.

The behavioral data we collected showed a significant NoGo specific reduction in response accuracy for HTA students compared to LTA students, which when taking into account similar findings from previous literature, can be assumed to support the notion that impaired executive functions in anxiety may be specific to inhibitory control and any domain general effect (Pacheco-Unguetti et al., 2012).

There's a great inconsistency in results from this research area in deep need of clarification since we haven't yet been able to reach a conclusion on whether there even is a relationship between TA and inhibitory cognitive control. Previous research using the same Go/NoGo paradigm found differences in response accuracy between HTA and LTA (Sehlmeyer et al., 2010; Righi et al., 2009). This incongruence may be a result of differences the experimental designs. In our study NoGo trials were presented as rare events only accounting for 15% of the total trials whereas in Karchs' et al. (2007) study there was an equal amount of each stimuli. This difference may be significant because evidence exists suggesting that task difficulty increases as the ratio of rare stimuli decreases, resulting in a higher amount of false trigger errors (Young et al., 2017).

Our ERP results yielded no significant difference in N2 peak amplitude or latency between HTA and LTA across Go and NoGo trials. However, we did find the P3 peak amplitude in HTA students to be significantly greater than that of LTA in both Go and NoGo trials.

Our results directly contradict the results of Sehlmeyer et al., (2010) who found the opposite of an increased N2 peak amplitude in HTA but no difference in P3 peak amplitude, however this was one of the studies that found no behavioural difference between TA groups and had the variations experimental design as stated previously.

The N2 component is suggested to reflect pre-response impulse evaluation as a component of motor inhibition (Huster et al., 2013). Whilst our findings do not concur, pervious research has linked the reduces N2 amplitude and increased latency to weaker inhibitory control, these altered N2 characteristics have not only been found in people with HTA, but also other conditions with overlapping impairments to executive function in the form of inattentiveness and like ADHD (Woltering et al., 2013) and Depression (Katz et al., 2010).

The P3 is implicated with response monitoring and evaluation and would therefore be linked to reinforcement learning (). Our findings of an increased P3 amplitude in both Go and NoGo trials argues against the notion that cognitive impairments in anxiety a result of negatively biased appraisal since if that were the case, we would only expect there to be a difference in NoGo trials like was found in Xia et al., (2020). In addition, the P3 peak latency was higher in LTA compared to HTA for NoGo trials which contradicts our prediction.

In summary, executive function in anxiety does seem to exclusively effect NoGo accuracy as per our behavioural findings but based off ERP data we were unable to confirm whether this disproportionate impairment to NoGo performance was due to inhibitory associated N2 or P3

ERP components as we hypothesised. Confirming which component of executive function is affected may aid in developing more effective cognitive therapy strategies for anxiety. More research is needed in light of substantial conflicting evidence on both sides of the argument.

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